Operating Systems Week 2 Lecture 2

Junaid Abdul Wahid

Previous

- History of Operating System
- The Operating System Zoo

The Operating System Zoo

Mainframe operating systems

Room-size computers

High I/O capacity

Offers:

Batch processing (no interaction, such as large reports)

Transaction processing (large number of small requests)

Timesharing (multiple users sitting in front of clients)

The Operating System Zoo

Server operating systems

Offer services like print, file, or web

UNIX, Windows 2000, Linux

Multiprocessor operating systems

Parallel computing

Personal computer operating systems

Single user with a good GUI, such as Windows 98, Windows 2000, Macintosh OS, Linux

The Operating System Zoo

Real-time operating systems

E.g. industrial process control systems where each job must be completed in the specified time.

Hard real-time (nuclear reactor control systems) or soft-real time systems (e.g multimedia systems) depending on the acceptance of missing deadlines

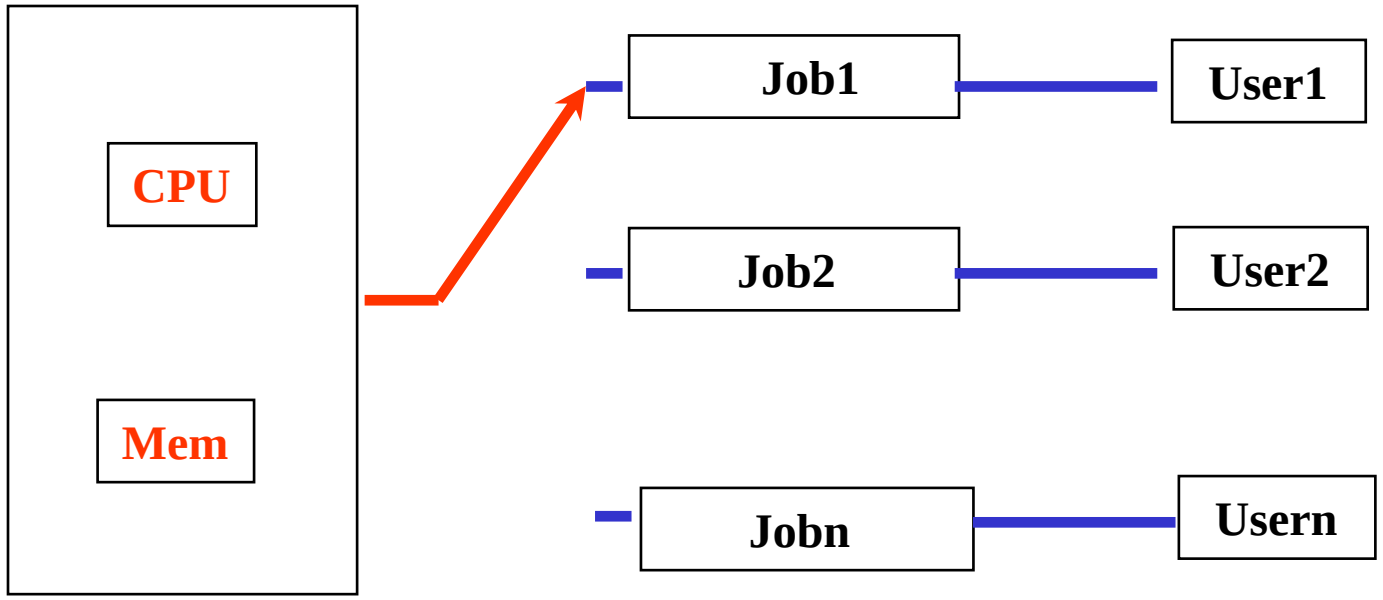
Embedded operating systems

Real-time systems with some resource constraints like memory, CPU, power.

Smart card operating systems

Extremely primitive OS running on credit card-sized devices with a CPU.

Time-Sharing Systems



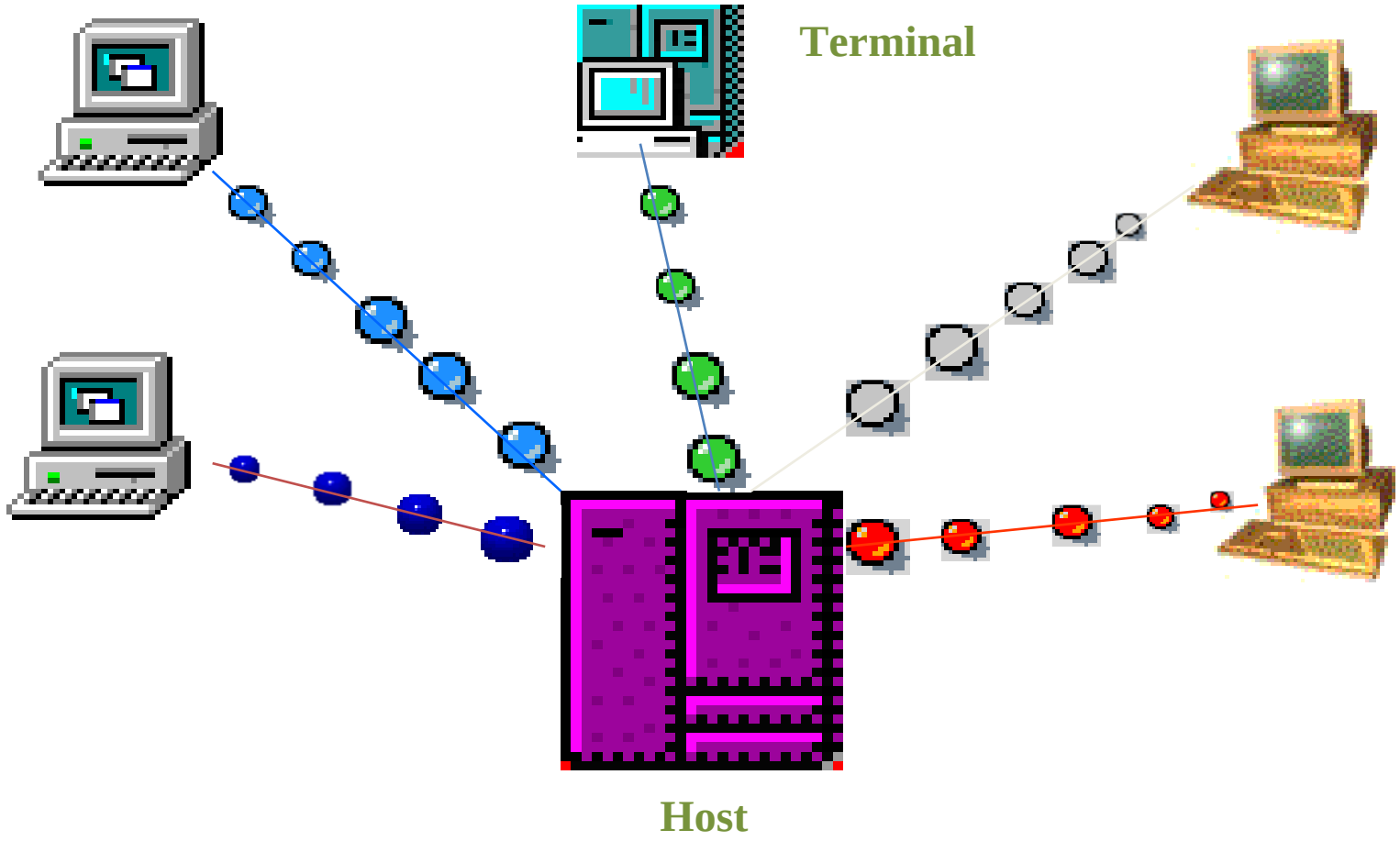
Time-Sharing Systems (Cont.)

The CPU is multiplexed among several jobs that are kept in memory and on disk (the CPU is allocated to a job only if the job is in memory).

A job is swapped in and out of memory to the disk. On-line communication between the user and the system is provided; when the operating system finishes the execution of one command, it seeks the next "control statement" not from a card reader, but rather from the user's keyboard.

On-line system must be available for users to access data and code.

Time-Sharing Systems (Cont.)



Real-Time Systems

Often used as a control device in a dedicated application such as controlling scientific experiments, medical imaging systems, industrial control systems, and some display systems.

Well-defined fixed-time constraints.

Strict time limits



Real-Time Systems (Cont.)



Real-Time Systems (Cont.)



Real-Time Systems (Cont.)



Real-Time Systems (Cont.)

Hard real-time system

Secondary storage limited or absent, data stored in short-term memory, or read-only memory (ROM)

Conflicts with time-sharing systems, not supported by general-purpose operating systems.

Soft real-time system

Limited utility in industrial control or robotics

Useful in applications (multimedia, virtual reality) requiring advanced operating-system features.

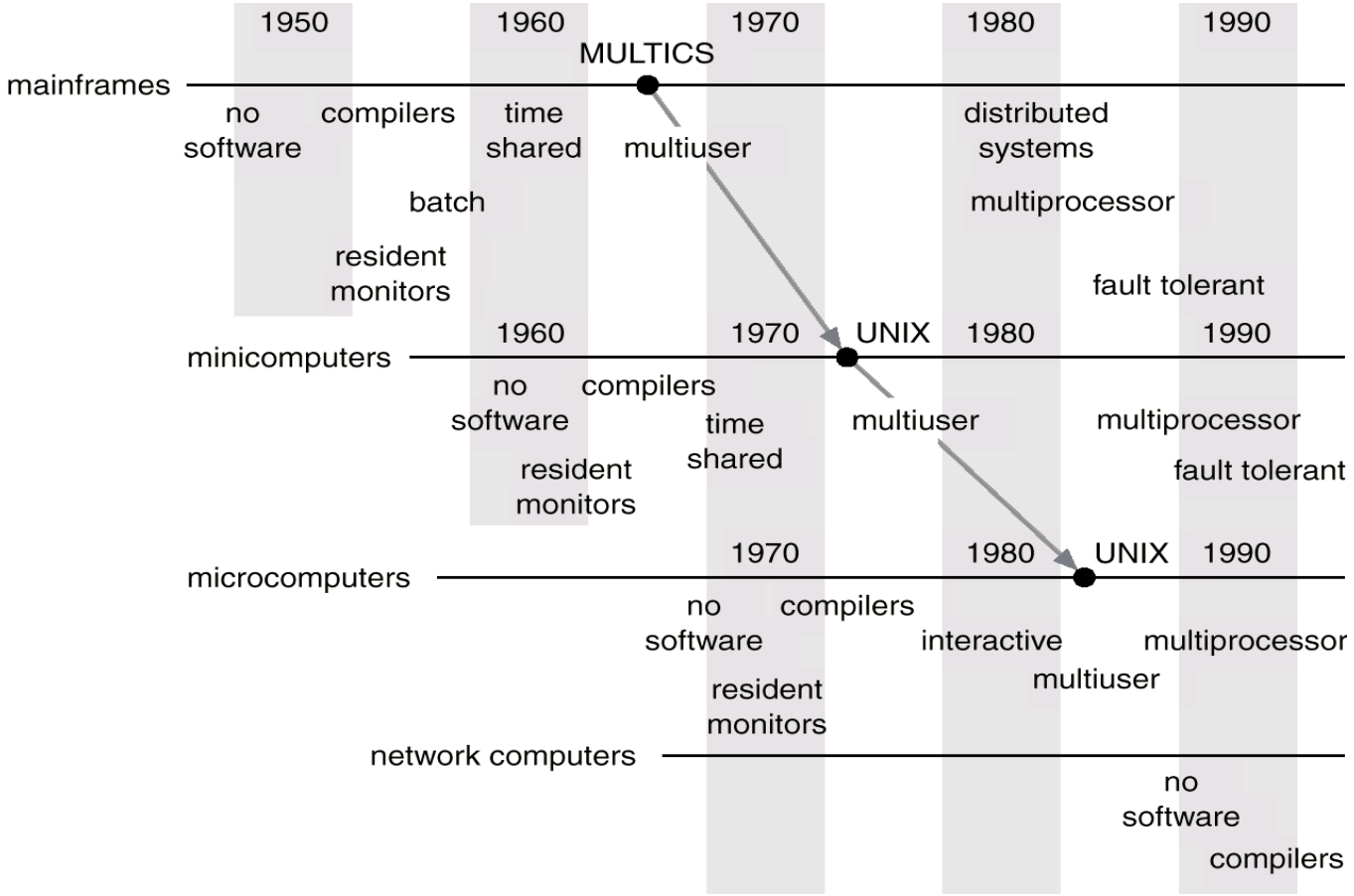
Personal-Computer Systems

Personal computers – **computer system dedicated to a single user.**
I/O devices – **keyboards, mice, display screens, small printers.**

User convenience and responsiveness.
Can adopt technology developed for larger operating system, often individuals have sole use of computer and do not need advanced CPU utilization of protection features.

It can be developed with large-scale operating systems, and personal exclusive computers do not require protection technology for advanced CPU use

Migration of Operating-System Concepts and Features



Parallel Systems

Multiprocessor systems with more than one CPU in close communication.

Tightly coupled system– processors share memory and a clock; communication usually takes place through the shared memory.

Advantages of parallel system:

- **Increased *throughput***
- **Economical**
- **Increased reliability**
  - **graceful degradation**
  - **fail-soft systems**

Parallel Systems (Cont.)

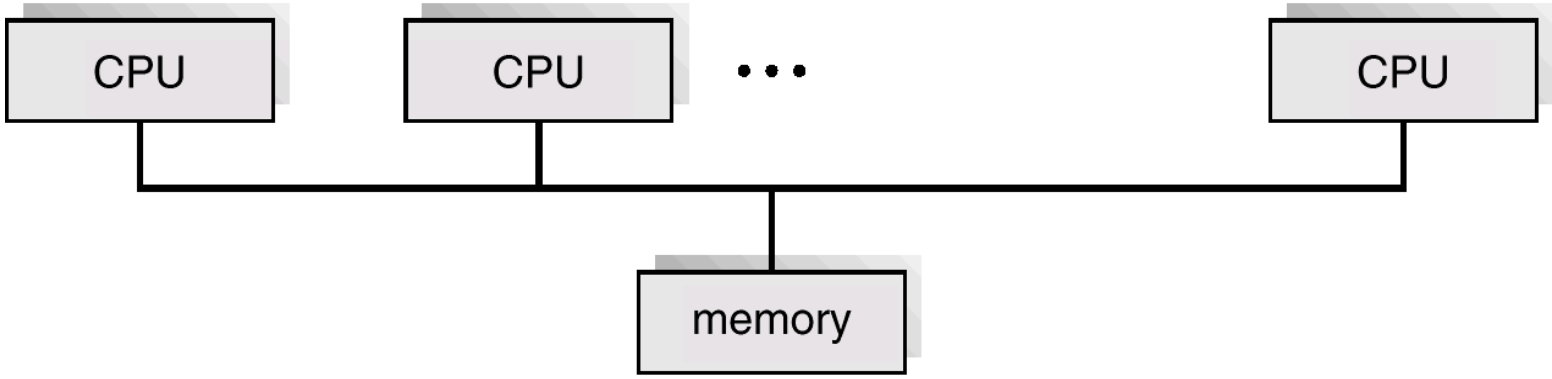
Symmetric multiprocessing (SMP)

- **Each processor runs an identical copy of the operating system.**
- **Many processes can run at once without performance deterioration.**
- **Most modern operating systems support SMP**

Asymmetric multiprocessing

- **Each processor is assigned a specific task; master processor schedules and allocates work to slave processors.**
- **More common in extremely large systems**

Symmetric Multiprocessing Architecture



Distributed Systems

Distribute the computation among several physical processors.

Loosely coupled system – each processor has its own local memory; processors communicate with one another through various communications lines, such as high-speed buses or telephone lines.

Advantages of distributed systems.

- **Resources Sharing**
- **Computation speed up**
- **Reliability**
- **Communications**

Distributed Systems (Cont.)

Network Operating System

- provides file sharing
- provides communication scheme
- runs independently from other computers on the network

Distributed Operating System

- less autonomy between computers
- gives the impression there is a single operating system controlling the network.

Types, Functions

Types of OS

- **Single user single task: CP/M, MS-DOS**
- **Single user multiple tasks: OS/2, WINDOWS**
- **Multiple users multiple tasks: UNIX**

Main Functions of OS

- **CPU management**
- **Memory management**
- **File management**
- **Device management**
- ***Job management***

Design and Construction

Concurrence
Sharing
Virtual
Asynchrony

Concurrence

Concurrence& Parallel

Concurrence: two or more events occur at the same time interval.

Parallel: two or more times occur at a unified time.

Sharing

- Resources in the system can be used by multiple concurrent processes in the memory.
- There are two ways to share resources:
  - mutual exclusion sharing
  - simultaneous access.

Sharing Contd.

Mutual exclusion and sharing

some resources can be used by multiple processes, but can only be occupied by one process within a specific period of time. Such resources are called critical resources; other processes that want to use the resource must wait for the current process to release the resource.

Simultaneous access

Sharing

Mutual exclusion and sharing

Simultaneous access

some resources (such as disks) can allow multiple processes to access simultaneously. Note that the "simultaneous" here is a macro concept, and at the micro level, it is often these processes that access resources alternately.

Virtual

Virtual: to change a physical entity into several logical counterparts by some kind of technology.

Physical entities actually exist, while the latter is virtual and what users feel.

For example, there is only one CPU in a multi-channel time-sharing system, but each end user thinks that there is one CPU specifically serving it, this is to use the multi-channel program technology to convert a physical CPU into a multi-logic CPU.

Operating System Services

- In view of resource management, the functionalities of operating systems
  - process management
  - memory management
  - storage management
  - I/O management
- With respect to man-machine interaction,
  - protection
  - security

Operating System Services

Services

services can be viewed as the functions that OS provides for its users, or as the OS capabilities supporting programs executing in computer systems. ---what

services are implemented through executing of , and cooperation among, the components of OS

Several types of user-oriented services

user interfaces

command-line interface

batch interface (off-line interface)

graphical user interface (GUI)

Operating System Services (cont.)

program execution – system capability to load a program into memory and to run it

- related to *process management* and *memory management* components

I/O operations – since user programs cannot execute I/O operations directly, the operating system must provide some means to perform I/O

- related to *I/O system management components*

file-system manipulation – capability to read, write, create and delete files

- related to *file management*, *secondary storage management* and *I/O system management* components

Operating System Services (cont.)

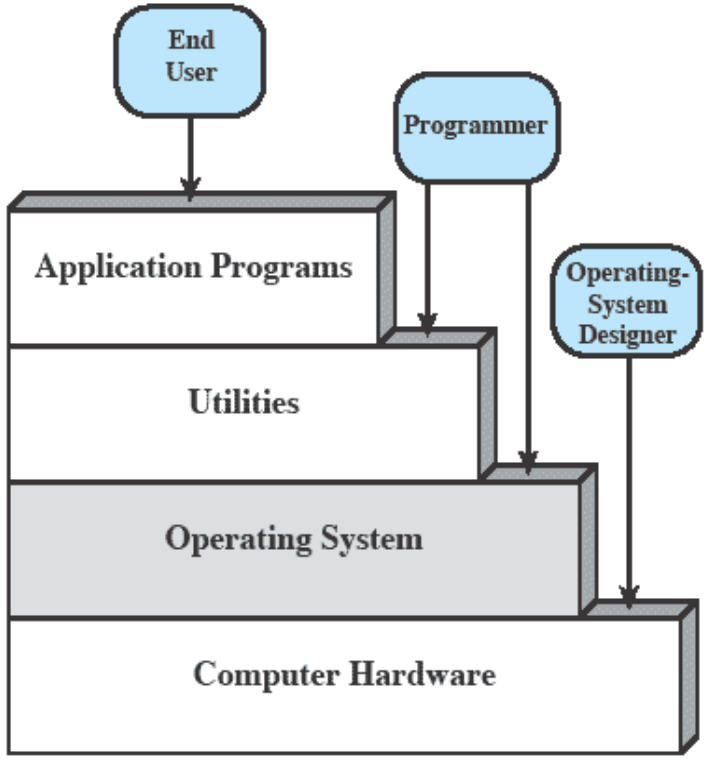
- communications – exchange of information between
  - ***processes*** executing either on the same computer or on different systems tied together by a network
    - implemented via *shared memory* or *message passing*
    - related to *networking* or *I/O system management* components
- error detection – ensure correct computing by detecting errors in the CPU and memory hardware, in I/O devices or in user programs
    - e.g. booting

Operating System Services (cont.)

Additional functions exist not for helping the user, but rather for ensuring efficient system operations

- **resource allocation** – allocating resources to multiple users or multiple jobs running at the same time
- accounting – keep track of and record which users use how much and what kinds of computer resources for account billing or for accumulating usage statistics
- protection and security– ensuring that all access to system resources is controlled.

Layers and Views



Services Provided by the Operating System

- Program development
  - Editors and debuggers.
- Program execution
  - OS handles scheduling of numerous tasks required to execute a program
- Access I/O devices
  - Each device will have unique interface
  - OS presents standard interface to users

Services cont…

- Controlled access to files
  - Accessing different media but presenting a common interface to users
  - Provides protection in multi-access systems
- System access
  - Controls access to the system and its resources

Services cont…

- Error detection and response
  - Internal and external hardware errors
  - Software errors
  - Operating system cannot grant request of application
- Accounting
  - Collect usage statistics
  - Monitor performance

The Role of an OS

A computer is a set of resources for the movement, storage, and processing of data.
The OS is responsible for managing these resources.